

TuniCars+

Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

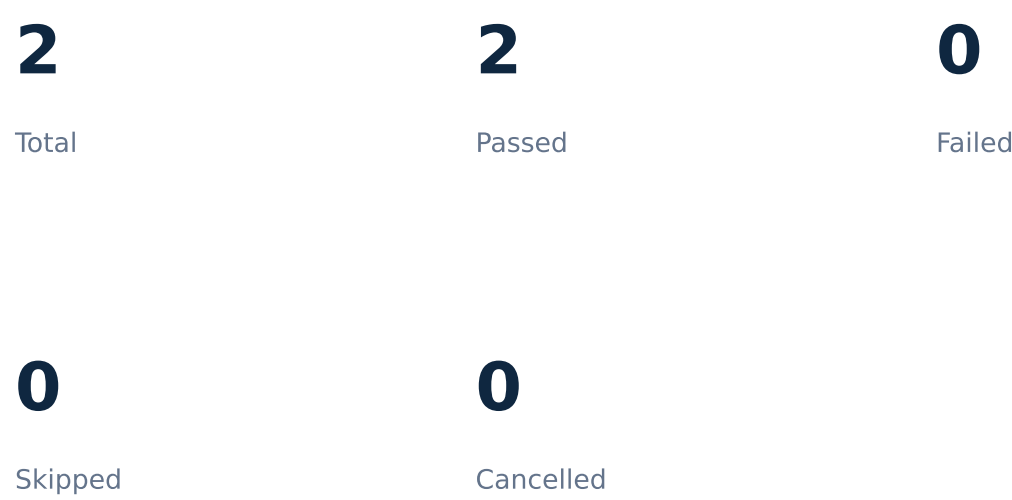
20260813_011353_cef352

Date

2026-08-13T00:13:53.338749+00:00

Executive Summary

20260813_011353_cef352



Overall measured availability: 7.08%

During this campaign, 2 of 2 scenarios passed, 0 failed and 0 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	kubernetes	swarm
container-kill	1	1

Full Results Table

Scenario	Platform	Status	Detection (s)	Recovery (s)	Average Response (ms)	P95 Response (ms)	Availability (%)	Recovery Success
container-kill	swarm	PASS	4.36	20.547	10.307	17.912	9.52	True
container-kill	kubernetes	PASS	58.188	123.844	1905.458	2075.473	4.65	True

swarm — container-kill

State: **Pass**
Detection Time (s): 4.36

Recovery Time (s): 20.547

HTTP Success: 2

HTTP Failures: 19

Availability (%): 9.52

CPU (%): 90.1

Memory (MB): 7678.12

Recorded interpretation

docker rm CLI success is distinct from the expected forced backend container termination (commonly exit code 137).

kubernetes — container-kill

State: PASS
Detection Time (s): 58.188

Recovery Time (s): 123.844

HTTP Success: 2

HTTP Failures: 41

Availability (%): 4.65

CPU (%): 73.7

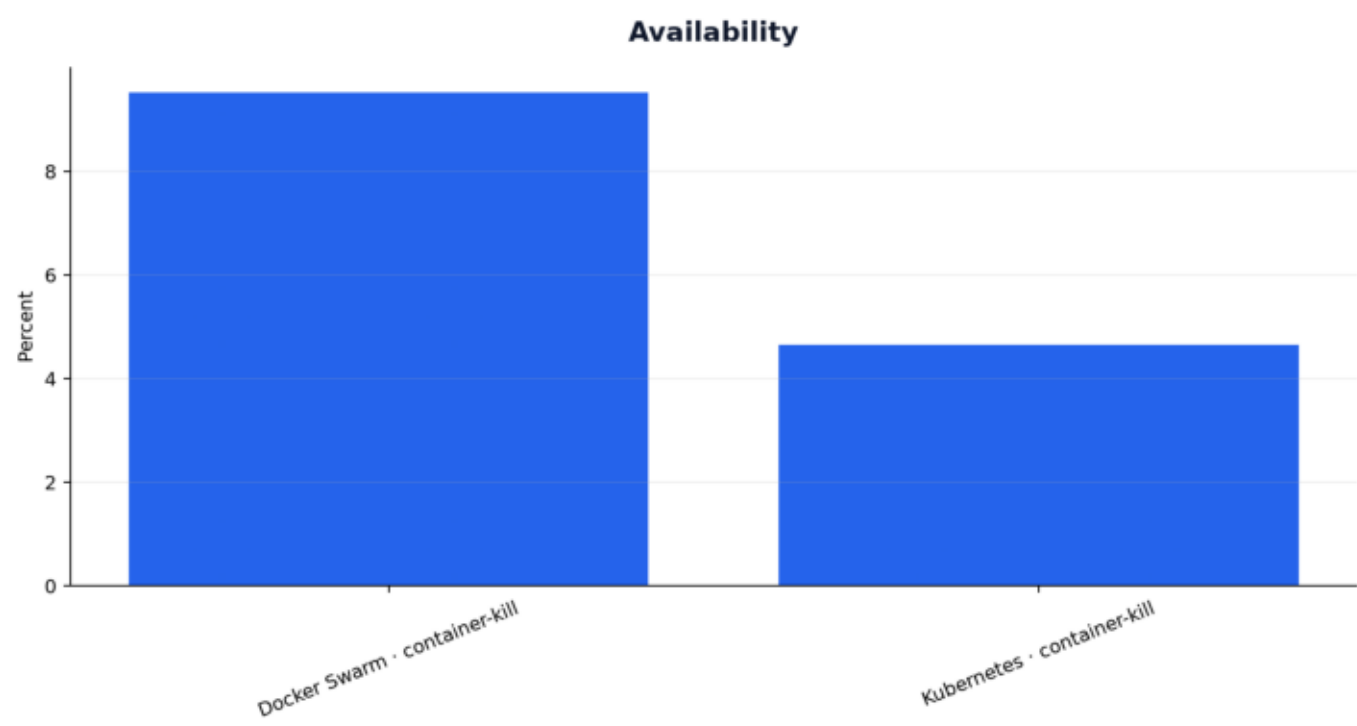
Memory (MB): 7616.57

Recorded interpretation

No error or limitation recorded.

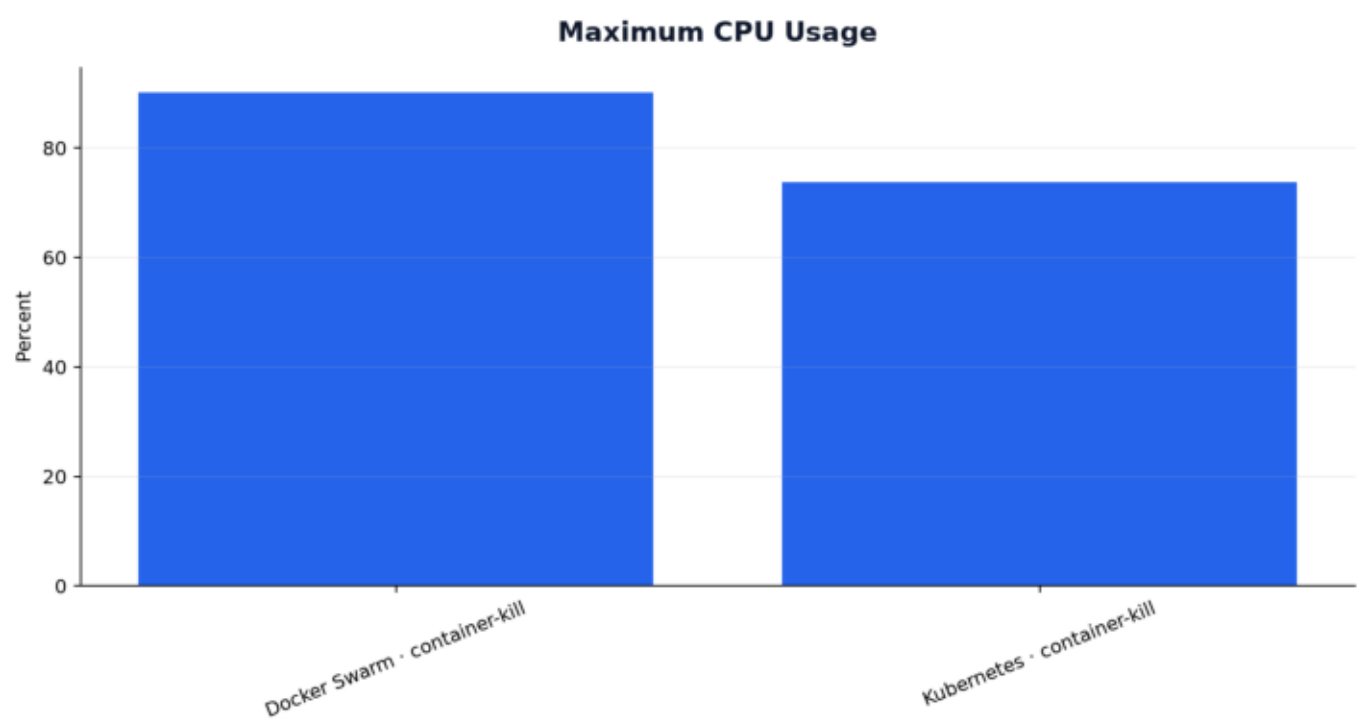
Availability

Current-session measured data



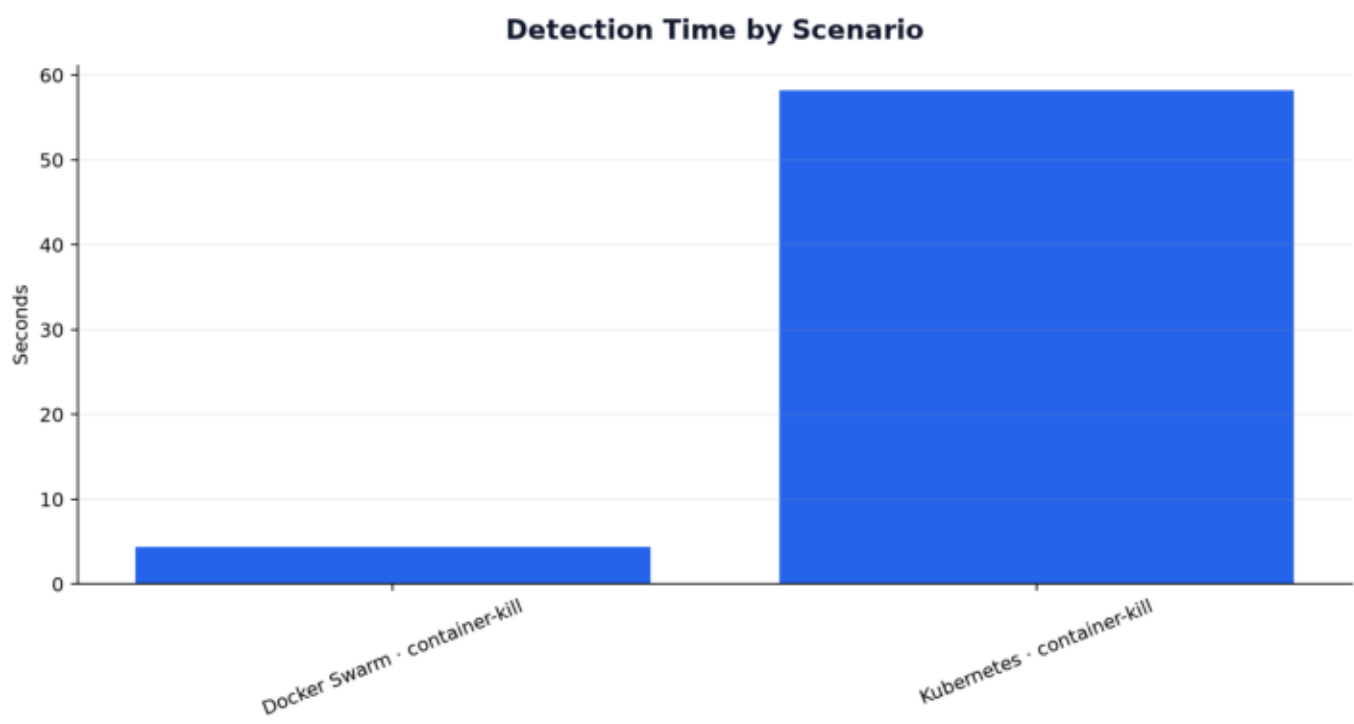
Cpu Usage

Current-session measured data



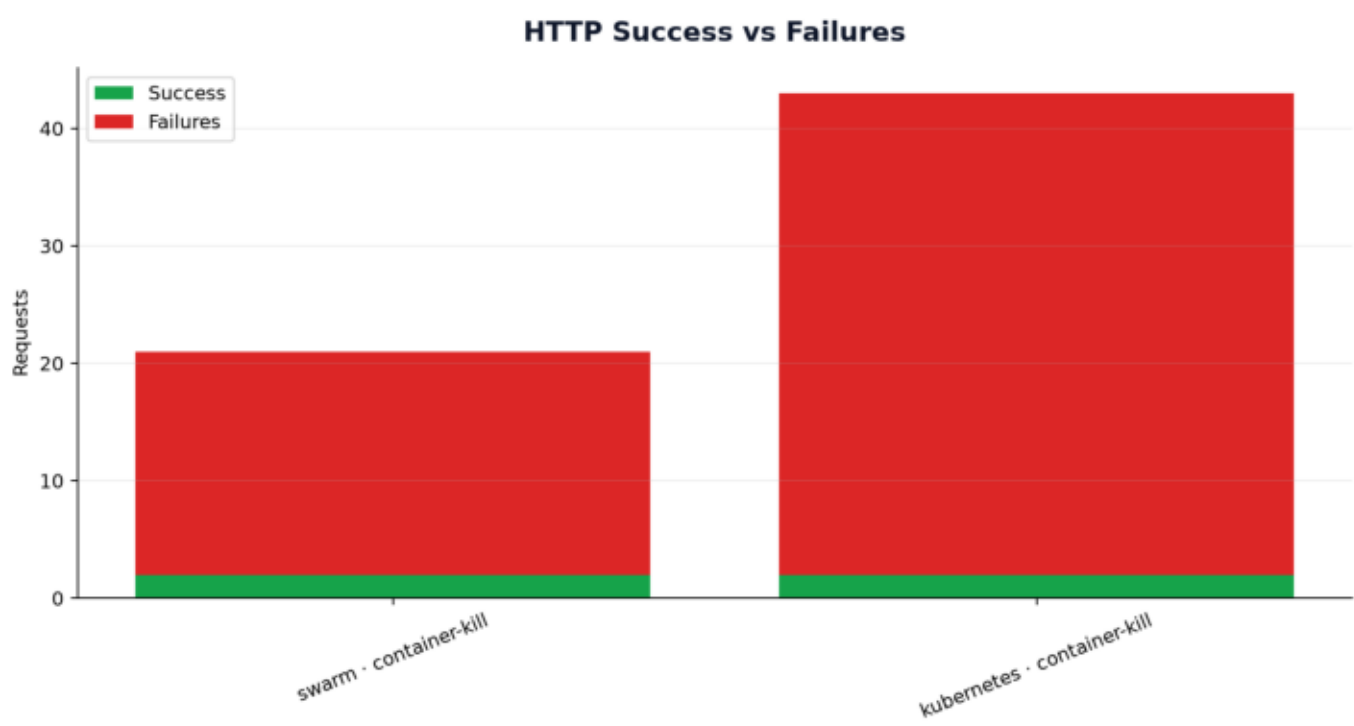
Detection Time

Current-session measured data



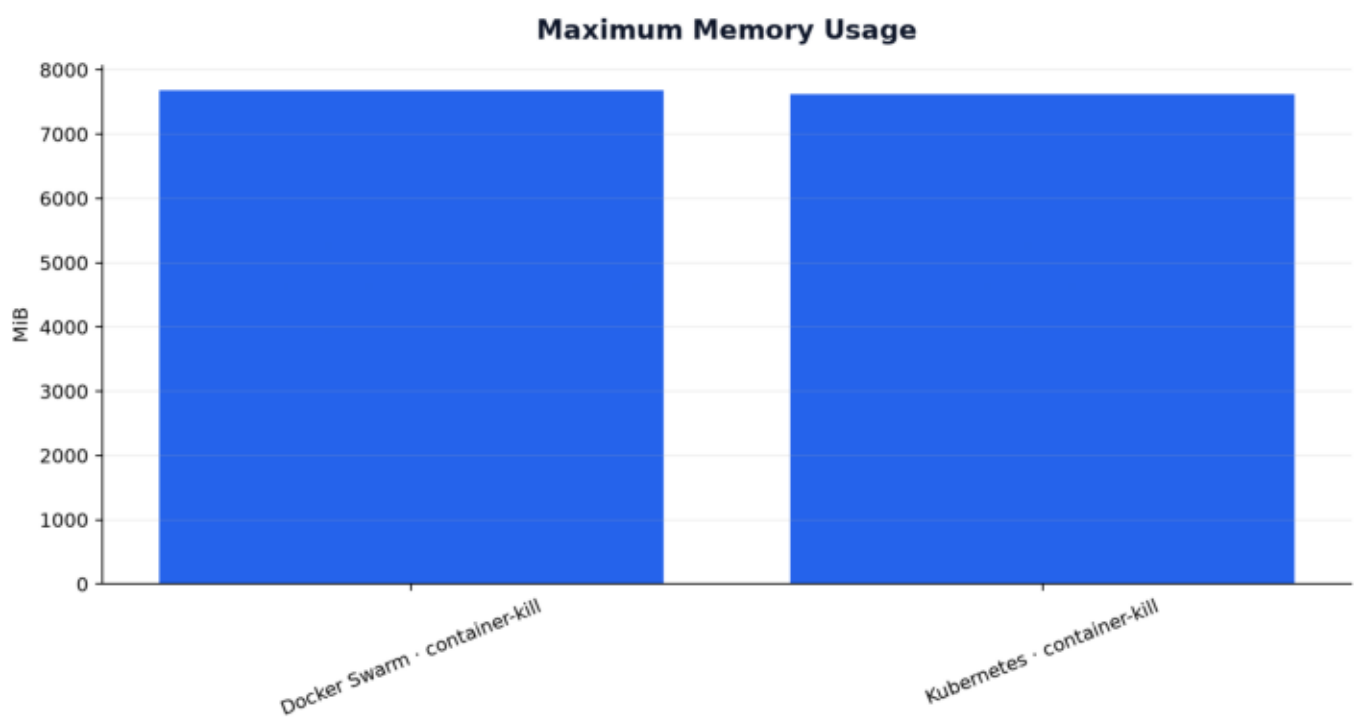
Http Requests

Current-session measured data



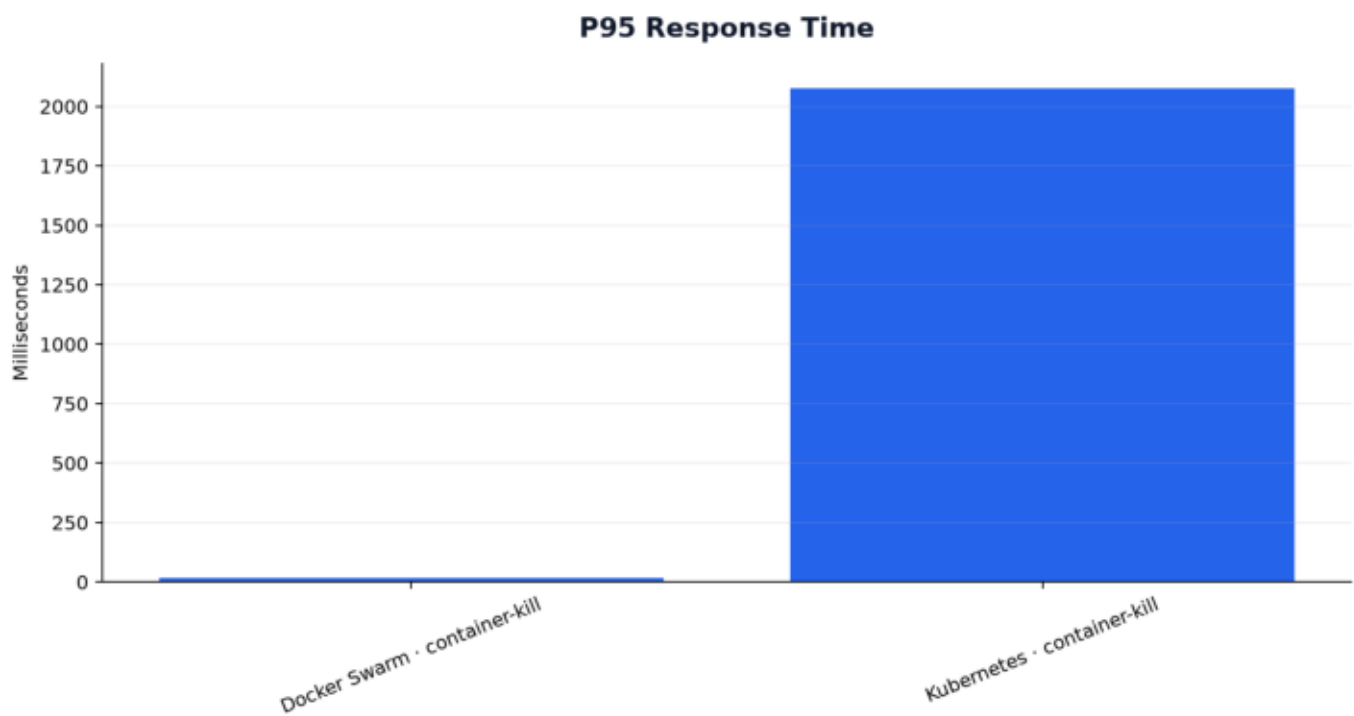
Memory Usage

Current-session measured data



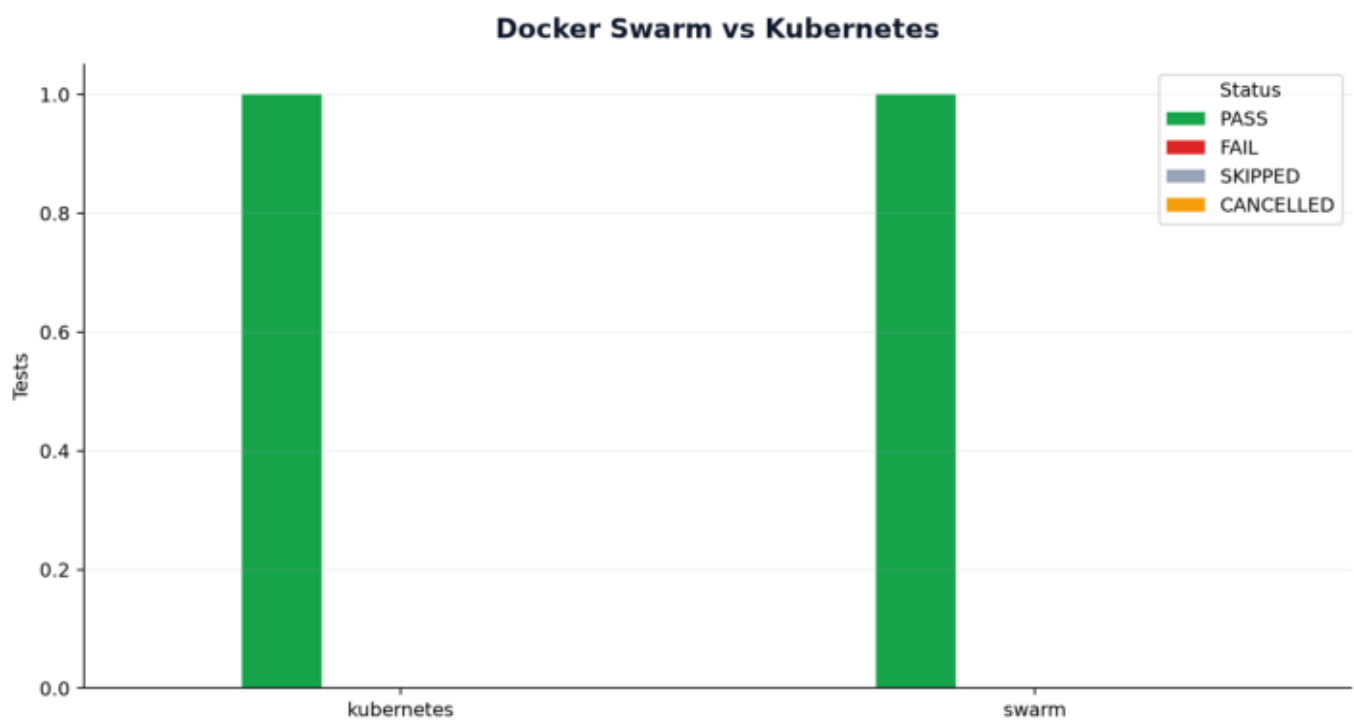
P95 Response Time

Current-session measured data



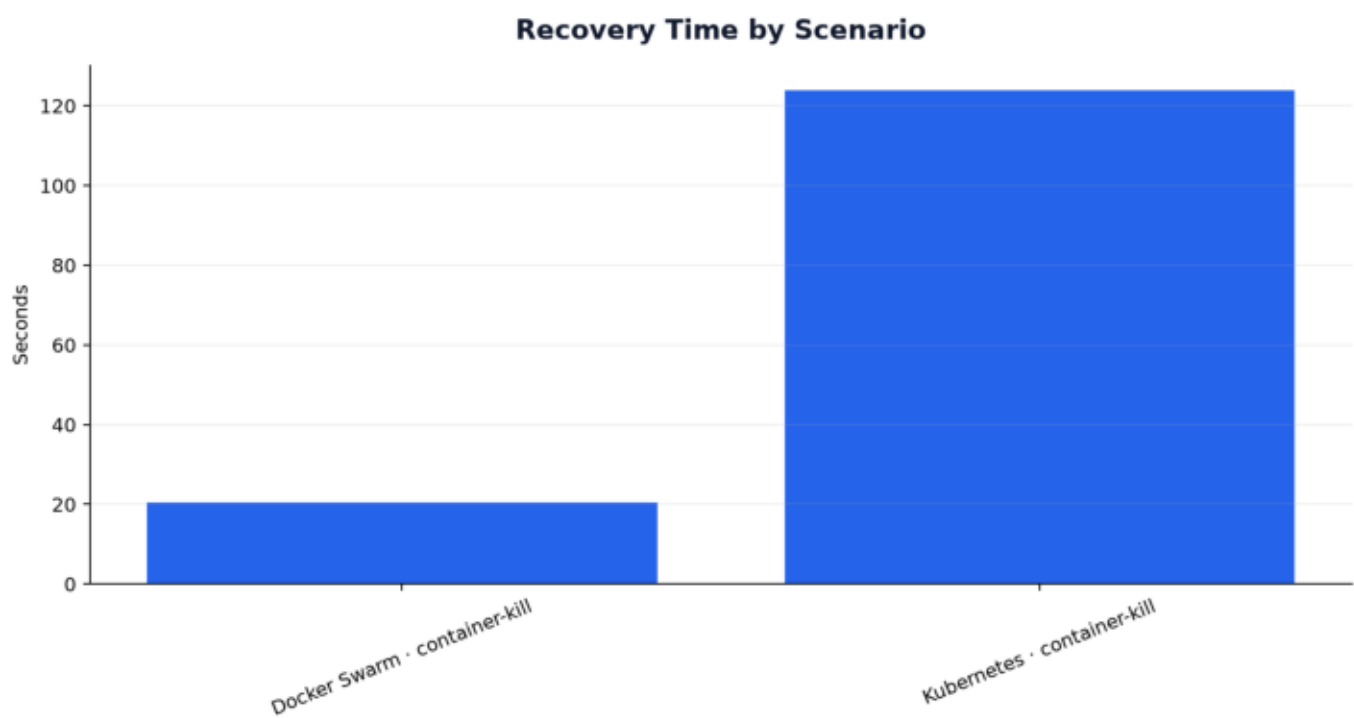
Platform Comparison

Current-session measured data



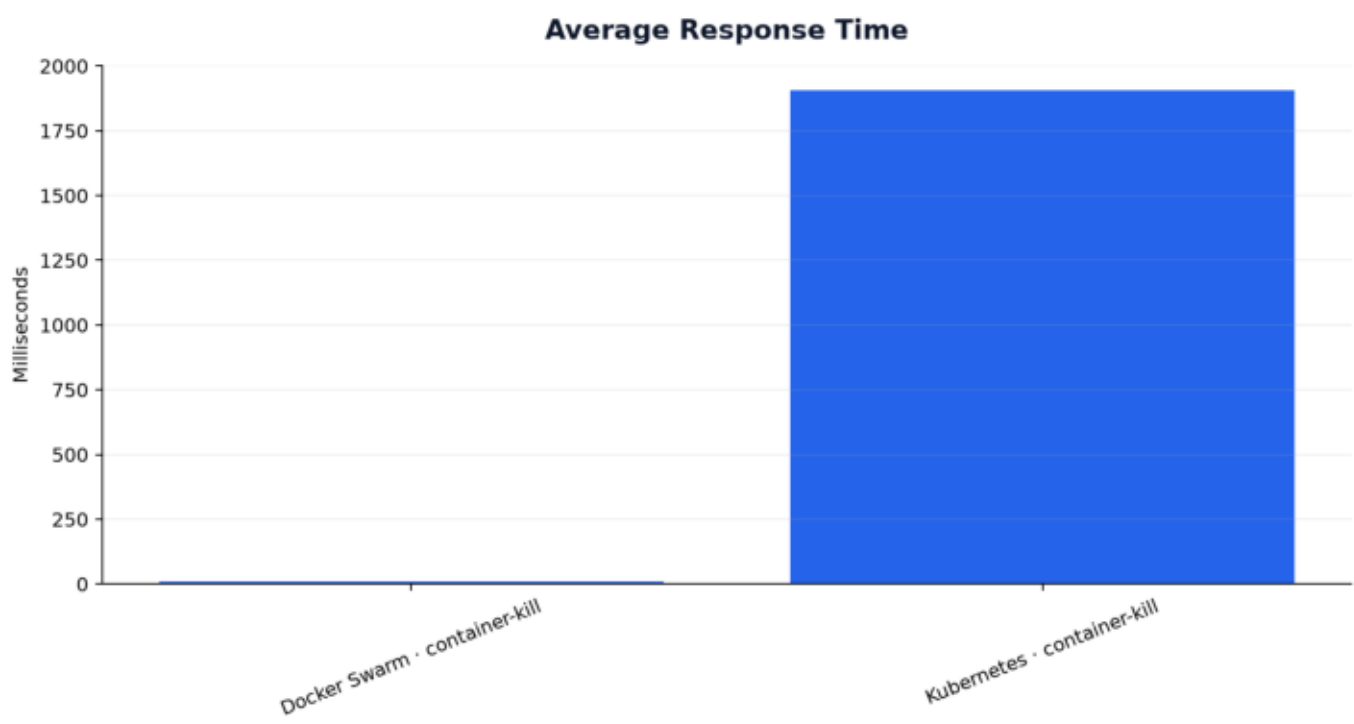
Recovery Time

Current-session measured data



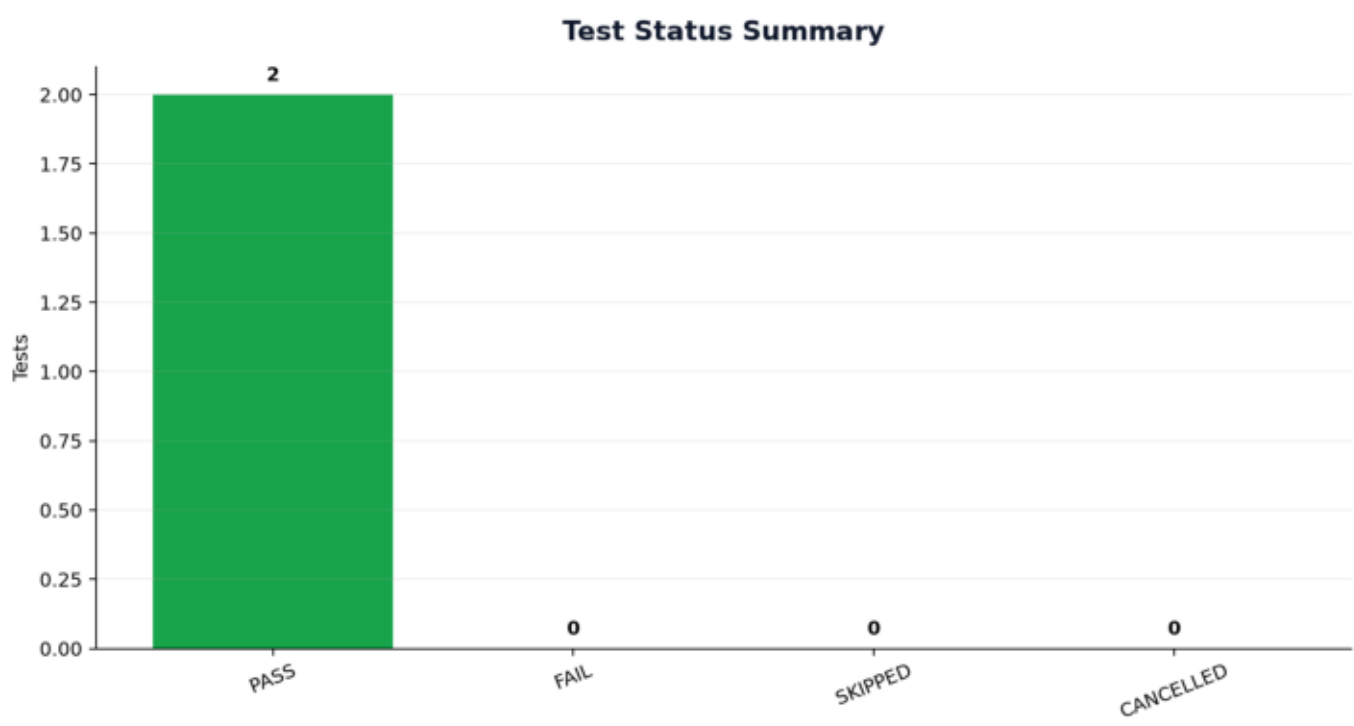
Response Time

Current-session measured data



Status Summary

Current-session measured data



Errors and Limitations

Platform	Scenario	Status	Error	Limitation
—	—	—	No errors recorded.	No limitations recorded.

Final Conclusion

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During this campaign, 2 of 2 scenarios passed, 0 failed and 0 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.